

# #11 Increasing Reward Quality & Density for Verifiable Rewards

Tristan Till

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Chair of Software Engineering for Business Information Systems (sebis)
Department of Computer Science
School of Computation, Information and Technology (CIT)
Technical University of Munich (TUM)
https://www.cs.cit.tum.de/sebis

# Any Questions?

- Course organization
- Project
- Anything else


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# Motivation

- **Analytics**
  - **Sparse rewards** are a known problem in reinforcement learning
  - We are interested in identifying, which steps make one agent successful and another fail (=*“critical”*)
  - No notion of what makes a state *objectively difficult* (human bias not a reliable proxy)
  - Use a **weaker version of the same model** for comparison!

- **Intervention**
  - Isolate states that are **learnable** (not noise) but **difficult** (observer disagreement) for the model at this moment in time
  - Minimize divergence from already learnt policy (*easy states*)
  - Manipulate gradients to convert difficult into easy states!


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# Key Questions

- *How to generate a weaker version of the same model?*
  - *What constitutes as true “weakness”?*
  - *How can this property be reduced reliably?*
  - *How to ensure this method remains on-policy?*
  - *How to verify that this makes the most sense beyond all other potential methods?*
- *How to maintain routine behavior, while training variable one?*
  - *How to ensure divergence of “easy” states is bounded?*
  - *How to differentiate noise from difficulty reliably?*
  - *How to verify that this actually does what it is supposed to?*
  - *To what extent does this improve downstream performance?*


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# Approach

- Slim/targeted hypothesis
  - *E.g., PPO instead of general RL; singular value pruning via tensor disentanglement*
  - *Use some theory to ground this: related work, math, …*
- Broad coverage of experiments
  - *Methodological soundness >> impact (objectively analyze your hypothesis)*
  - *How effective is your method empirically when compared with theory*
- Small subset of targeted insights
  - *Only makes sense when soundness is guaranteed*
  - *Answer three questions under the hypothesis “If our method is sound, we could say…”*


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# Discussion 24.03.

- *How do we get the “smaller model”?*
  - Use the “tensor disentanglement” code provided by me, focus on one LLM for now and determine its disentangled form
  - This would already be a major contribution for your seminar paper

- *How does this relate to RL?*
  - Treat the initial phase as inference-only evaluation, in which each generated token represents one action along a hypothetical RL trajectory
  - The previously generated text is our “state”, each predicted token our next “action”

- *What exactly do we analyze?*
  - 1) Let the large model act and measure the level of disagreement at different pruning levels of the smaller model (*“observer pattern”*) to answer for how many states is non-standard behavior relevant.
  - 2) Let the smaller model act and measure the downstream performance under pruned behavior to answer at what point disagreement in paths measurably reduces reward


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